# LUKE HOJNICKI

Chicago, IL | 402-547-9925 | lhojnick@purdue.edu | linkedin.com/in/luke-hojnicki

## EDUCATION

**Purdue University** West Lafayette, IN

*B.S. Electrical Engineering, Microelectronics & Semiconductors Concentration* *Expected May 2028*

- GPA: 3.7/4.0, Dean's List and Semester Honors every semester
- Scholarships: Joseph Geisinger Scholarship, Conagra Brands Foundation Scholarship
- Study Abroad: Universidad Carlos III de Madrid (CIEE), Madrid, Spain, Spring 2027

## EXPERIENCE

**International Motors** Chicago, IL

*Research & Development Intern, Emissions Compliance Data Science* *May 2026 – Aug. 2026*

- Built an AI triage model (XGBoost + LLM, Databricks) projected to halve review time for 10,000 emissions warranty claims via a Power BI dashboard; secured approval as the first project through the company's new AI governance process
- Won company-wide AI hackathon (1st of ~50 teams, mostly full-time engineers) with a predictive maintenance tool; pitched to executives and was invited to parent company TRATON Group's international hackathon
- Benchmarked the top-selling midsize engine against Cummins across 800 vehicle configurations, showing it beat Cummins on fuel efficiency in the high-volume configurations that outsell the rest roughly 100 to 1

**VIP Semiconductors, Purdue Birck Nanotechnology Center** West Lafayette, IN

*Undergraduate Researcher, Statistical Process Control Etch Team* *Aug. 2026 – Present*

- Diagnosing why thin-film circuits miss electrical specs on an 8-person team, using statistical process control to isolate whether defects stem from wafer-to-wafer drift or within-wafer variation in a plasma etch step
- Own metal deposition for the team (6 wafers fabricated in 2 months), brief 12 faculty and research scientists weekly, and co-authored the team's abstract for Purdue's undergraduate research symposium (poster, Dec. 2026)

**Purdue Coherent and Quantum Optics Lab** West Lafayette, IN

*Undergraduate Researcher, Electronics Design, Dr. Daniel Elliott* *Aug. 2025 – May 2026*

- Designed and built an 8-sensor magnetometer array end to end (PCB, firmware, Python interface), now used by the lab to measure magnetic fields in its cesium laser experiments and documented in a technical report for future students

**Connelly Electric** Chicago, IL

*Electrical Design Engineering Intern* *May 2025 – Aug. 2025*

- Resolved design conflicts across 8 trades on $10M+ industrial projects using 3D building models, catching clashes early
- Launched VR inspection of prefab assemblies before shipment to job sites, reducing rework in the fabrication shop

## LEADERSHIP

**Purdue ECE Ambassadors** West Lafayette, IN

*Executive Board, Professional Development Chair* *Jan. 2026 – Present*

- Co-led Purdue ECE's inaugural Leadership Summit (195 registrants, $5K budget); rebuilt the budget, recruited alumni support, and cut scope from four tracks to three when partner commitments stalled
- Co-recruited speakers from 6 firms, including Microsoft, Marvell, Northrop Grumman, Sandia, and Caterpillar
- Led a 7-person committee to run a career fair prep night for 150 ECE undergraduates

**Theta Chi Fraternity** West Lafayette, IN

*Mom and Dad's Day Chair* *Nov. 2025 – Present*

- Ran Mom's Day and Dad's Day events for 75 families each, raising $7,000 for the USO through a raffle and donations

## SKILLS & ADDITIONAL

**Data & AI:** Python, Excel, Power BI, Databricks, XGBoost, LLM APIs, MATLAB, C
**Analytical Methods:** Statistical Process Control, Design for Six Sigma (DMADV), Statistical Modeling
**Technical:** Cleanroom Microfabrication, Photolithography, E-beam Evaporation, ALD, PCB Design
**Interests:** Golf (6.2 HCP), Backcountry Skiing, Fly Fishing, The Beatles